SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – JIGSAW ACTIVITY

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO4 – Articulation (BL3)	**Assessment:**	Assessment Tool 2 – Jigsaw Activity
Weightage:	35% of CIA	**Total Marks:**	100
CO4 Statement:	*Apply hashing strategies for efficient data storage and sorting algorithms for arrangement of data.*		
Student Name:	N.S.V. Trivikram	**Reg. No.:**	192525063
Section / Batch:	AI and ML / 2025	**Date:**	21/08/2024

Code of Conduct

I N.S.V Trivikram (192524023)______________________________(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Trivikram________

Instructions

- Home Groups (8): 5 students, Expert Groups: One expert per topic. Total: 100 Marks.
- Students first discuss in Expert Groups. They return to Home Groups and teach teammates.
- Each student teaches Concept, Working, Advantages, Disadvantages, Applications, Time Complexity.

Section / Topic	Focus	Q Nos	Marks
Hashing & Sorting	Hashing, Resolution Techniques, Sorting	8 Teams	100
	GRAND TOTAL:	**100 Marks**	

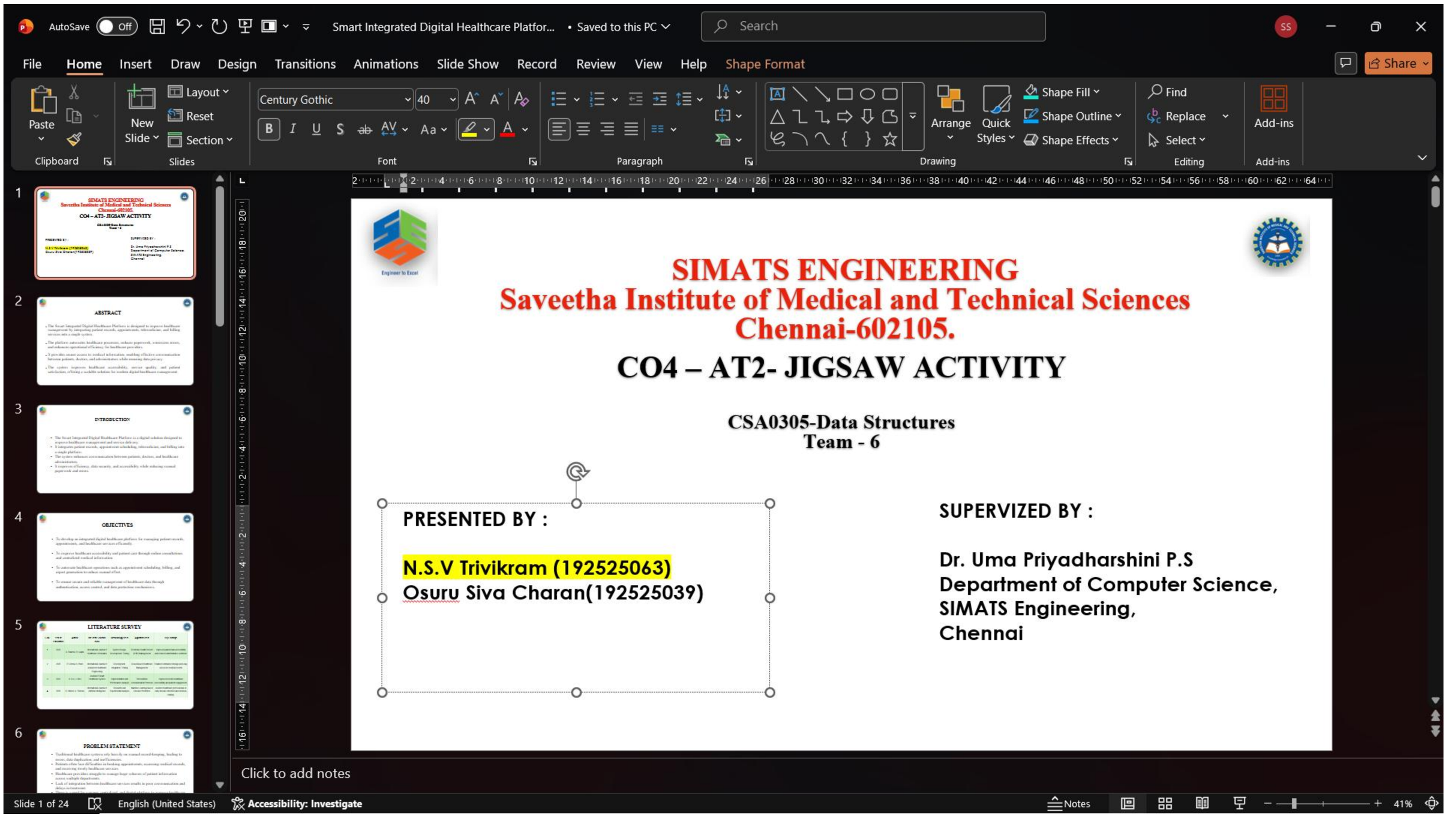
Annexure A – Jigsaw Activity

1. Scenario 1 – Supermarket Billing System (A supermarket stores 50,000 product IDs, Billing takes too long.)
    a. Suitable Hash Function
    b. Collision Resolution
    c. Sorting Method for sales reports
2. Scenario 2 – College Library Management (The library contains 150,000 books)
    a. Searching strategy
    b. Hashing method
    c. Sorting books by title
3. Scenario 3 – Online Banking System (Millions of account numbers are stored)
    a. Hash Function
    b. Collision handling
    c. Sorting transactions
4. Scenario 4 – Google Search Index (Google indexes billions of webpages)
    a. Hashing strategy
    b. Collision handling
    c. Sorting search results
5. Scenario 5 – IPL Scoreboard (Every ball updates scores)
    a. Hash Function
    b. Collision handling
    c. Fast Sorting
6. Scenario 6 – Aadhaar Citizen Database (The Government stores over 1.4 billion Aadhaar records. Citizens frequently search their details using Aadhaar numbers. Duplicate entries must be avoided, and reports need to be generated in ascending district order.)
7. Scenario 7 – Food Delivery Application (A food delivery platform manages 100,000 restaurants and 1 million customer orders. Customers search restaurants by name while orders are sorted according to delivery time.)
8. Scenario 8 – Cricket Tournament Statistics (A cricket analytics company stores player statistics after every match. Coaches need quick retrieval of player data and ranked scorecards after each game.)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Accurately interprets all problem requirements and constraints before coding.	Understands most requirements with minor omissions.	Partially understands the problem; misses some constraints.	Misinterprets the problem or ignores key requirements.
Correctness of Solution	30	Program compiles successfully and passes all hidden and visible test cases.	Program passes most test cases with minor errors.	Program passes some test cases but has logical issues.	Program fails to compile or produces incorrect results.
Data Structure & Algorithm Selection	20	Selects the most appropriate data structure and algorithm with an optimized solution.	Chooses an appropriate solution with minor inefficiencies.	Uses a workable but less efficient approach.	Uses an inappropriate data structure or algorithm.
Code Quality & Documentation	20	Well-structured, modular, readable, and properly indented code with meaningful identifiers.	Readable code with minor formatting issues.	Basic coding style with inconsistent formatting.	Poorly organized code that is difficult to understand.
Time & Space Optimization	10	Demonstrates excellent optimization and correctly analyses time and space complexity.	Good optimization with minor inefficiencies.	Limited optimization and incomplete complexity analysis.	No optimization and incorrect or missing complexity analysis.

Faculty In-charge	Course Coordinator	Program Director
Signature: ______________	Signature: ______________	Signature: ______________
Date: __________________	Date: __________________	Date: __________________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CO4 – AT2- JIGSAW ACTIVITY
CSA0305-Data Structures
Team - 6
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